KotaAgent: An agent submitted to the ANAC 2026 SCM league

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Abstract

This paper presents KotaAgent, an autonomous agent for the ANAC 2026 SCML Standard track. The agent enforces profit margins, uses delivery window constraints to reduce inventory risk, and applies fallback purchasing to avoid production failures when profitable trades are impossible. Buy and sell price thresholds are independently adapted using exponential moving average (EMA) and relaxed after negotiation failures.

1 Introduction

In the ANAC SCML Standard track, agents manage factories by negotiating contracts with suppliers and consumers and scheduling production over a fixed horizon. The main challenges are inventory costs and penalties for failing to satisfy sales contracts.

A difficult case occurs when supplier prices exceed consumer prices, making profitable trades impossible. KotaAgent addresses this using profit margin constraints, fallback purchasing, delivery window control, and adaptive price thresholds.

2 The Design of KotaAgent

KotaAgent extends `StdSyncAgent` and negotiates with all partners simultaneously using `first_proposals()` and `counter_all()`.

2.1 Negotiation Strategy

Profit margin enforcement. The agent calculates acceptable buy and sell price ranges. Purchases are limited by a price ceiling, while sales require a price floor that covers input costs. Offers outside these ranges are rejected.

Adaptive price fraction. KotaAgent maintains independent buy and sell fractions, initialized to 0.5 and limited to [0.1,0.9]. These fractions determine the initial proposal position within acceptable ranges. During negotiation, acceptance thresholds gradually move toward the boundary as the deadline approaches.

EMA learning. After successful negotiations, the corresponding fraction is updated toward the agreed price position using EMA with weights 0.7 and 0.3. After failures, the fraction is increased by 0.05 to relax future proposals.

2.2 Concurrent Negotiation

The agent distributes required quantities among partners and tracks accepted quantities during `counter_all()`. Once sufficient contracts are obtained, additional offers are rejected to prevent over-contracting. Future deliveries are handled separately.

2.3 Risk Management

Delivery window constraint. Offers with delivery dates more than four steps ahead are rejected to reduce inventory risk.

Fallback buying. When profitable purchasing is impossible, KotaAgent accepts minimum supplier prices instead of stopping production. This avoids large shortfall penalties caused by zero production.

3 Evaluation

KotaAgent was evaluated against `SyncRandomStdAgent` using the `anac2026_std` tournament runner (20 steps, 50 configurations, 200 worlds).

Agent	Average Score
KotaAgent	**0.880**
SyncRandomStdAgent	0.818

Table 1: Average normalized scores.

Configuration	KotaAgent	SyncRandomStdAgent
Base	0.879	0.823
+ Delivery window	0.860	0.775
+ Separate fractions + failure learning	**0.910**	0.832
+ Fallback buying	0.880	0.818

Table 2: Incremental evaluation results.

KotaAgent outperformed the baseline in all configurations. Separate buy/sell adaptation achieved the best score. Fallback buying slightly reduced average profit but prevented severe failures in unfavorable worlds.

4 Lessons and Suggestions

Robustness matters. Strict profitability constraints can cause zero production, while fallback strategies avoid catastrophic losses.

Random baselines are insufficient. Adaptive strategies should be tested against diverse agent behaviors.

Independent adaptation improves performance. Separate buy and sell fractions allow better calibration than a shared parameter.

Conclusions

KotaAgent achieved robust negotiation performance by combining profit margin constraints, adaptive pricing, fallback purchasing, and delivery risk management. The proposed strategies improved both profitability and robustness against unfavorable market conditions.